SYLLABUS

FOR
FIRST YEAR COURSES

B.TECH. 1st YEAR

(Effective from 2025-26)

Estd. 2000

ABES ENGINEERING COLLEGE, GHAZIABAD

(An Autonomous College, Affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow)

ABES Engineering College, Ghaziabad
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Estd. 2000

B. Tech. First Year Evaluation Scheme (w.e.f: 2025-26)											
SEMESTER-I											
Category	**Subject Code**	**Subject**	**Periods**			**Internal Evaluation**		**External Evaluation**		**Total**	**Credits**
			L	**T**	**P**	**Theory**	**Practical**	**Theory**	**Practical**		
Basic Science Course /Engineering Science Course	25AS101/ 25ME101	Applied Physics/ Basics of Mechanical and Automation Engg.	2	1	0	30		70		100	3
Engineering Science Course	25EL101/ 25EC101	Basics of Electrical Engg./ Basics of Electronics Engg.	2	1	0	30		70		100	3
Basic Science Course	25AS102	Fundamentals of Linear Algebra and Statistics	2	1	0	30		70		100	3
Engineering Science Course	25CS101	Programming in C++	3	0	0	30		70		100	3
Basic Science Course/ Humanities and Social Science Course	25AS103/ 25HU101	Environment and Sustainability/Soft Skills	2	0	0	30		70		100	2
Engineering Science Course/ Humanities and Social Science Course	25AI101/ 25HU102	Essentials of AI/Design Thinking, Innovation and Entrepreneurship	2	0	0	30		70		100	2
Basic Science Course/ Humanities and Social Science Course	25AS151/ 25HU151	Physics Lab/ Professional Communication Lab	0	0	2		50		50	100	1
Engineering Science Course	25EL151/ 25EC151	Basic Electrical Engg. Lab/ Electronics Engg. Workshop	0	0	2		50		50	100	1
Engineering Science Course	25ME152/ 25ME151	CAD Lab/Digital Manufacturing Lab	0	0	2		50		50	100	1
Engineering Science Course	25VA151	Web Designing Workshop – I	0	1	2		50		50	100	2
Engineering Science Course	25CS151	Programming Lab in C++	0	0	2		50		50	100	1
Innovation, Entrepreneurship and Industry Connect for Employability Skills	25VA152	Holistic Skill & Innovation – I	0	0	2		100			100	1
Total			**13**	**4**	**12**					**1200**	**23**

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SEMESTER-II											
Category	**Subject Code**	**Subject**	**Periods**			**Internal Evaluation**		**External Evaluation**		**Total**	**Credits**
			L	**T**	**P**	**Theory**	**Practical**	**Theory**	**Practical**		
Basic Science Course /Engineering Science Course	25AS201/ 25ME201	Applied Physics/ Basics of Mechanical and Automation Engg.	2	1	0	30		70		100	3
Engineering Science Course	25EL201/ 25EC201	Basics of Electrical Engg./ Basics of Electronics Engg.	2	1	0	30		70		100	3
Basic Science Course	25AS202	Applied Calculus and Differential Equations	2	1	0	30		70		100	3
Engineering Science Course	25CS201	Data Structure using C++	3	0	0	30		70		100	3
Basic Science Course/ Humanities and Social Science Course	25AS203/ 25HU201	Environment and Sustainability/Soft Skills	2	0	0	30		70		100	2
Engineering Science Course/ Humanities and Social Science Course	25AI201/ 25HU202	Essentials of AI/Design Thinking, Innovation and Entrepreneurship	2	0	0	30		70		100	2
Basic Science Course/ Humanities and Social Science Course	25AS251/ 25HU251	Physics Lab/ Professional Communication Lab	0	0	2		50		50	100	1
Engineering Science Course	25EL251/ 25EC251	Basic Electrical Engg. Lab/ Electronics Engg. Workshop	0	0	2		50		50	100	1
Engineering Science Course	25ME252/ 25ME251	CAD Lab/Digital Manufacturing Lab	0	0	2		50		50	100	1
Engineering Science Course	25VA251	Web Designing Workshop – II	0	1	2		50		50	100	2
Engineering Science Course	25CS251	Data Structure Lab using C++	0	0	2		50		50	100	1
Innovation, Entrepreneurship and Industry Connect for Employability Skills	25VA252	Holistic Skill & Innovation – II	0	0	2		100			100	1
Total			**13**	**4**	**12**					**1200**	**23**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	I/II	Applied Physics	25AS101/ 25AS201	2	1	0	3

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
36	12	0	42	90

Course Objective (s)
This course aims to impart basic knowledge to engineering students with the concepts of quantum mechanics, storage & semiconductor devices, electromagnetic theory, laser and fibres for solving problems in quantum applications, communication, and modern technologies.

Syllabus

Unit No.	Topics	CI Hours
1	**Quantum Physics:** Need of Quantum mechanics, Wave-Particle Duality, Group and Phase velocity, wave function and its interpretation, Schrödinger equation (1D), Particle in a box (infinite potential well), Introduction to Quantum tunnelling, Quantum computing and cryptography.	8
2	**Emerging Physics for Engineering Applications:** Basics of Nanophysics, Quantum confinement in nanotechnology, 1D, 2D, 3D Nanomaterials, Fabrication techniques of nanomaterials: – Chemical Vapor Deposition, Sol-Gel, Applications of Nanotechnology- Quantum Dot, nanoscale memory, air and water purification, cosmetic industry. Superconductors: Meissner effect, temperature dependence of critical field, Type I & Type II superconductors. Introduction to Spintronics, Giant Magnetoresistance (GMR) for memory storage.	7
3	**Semiconductor & devices:** Classification of solids on the basis of energy band gap, types of semiconductor, Concept of Fermi level, position of fermi level in intrinsic and extrinsic semiconductors, mobility, effect of temperature on conductivity of semiconductor, Carrier generation/ recombination phenomena, Formation of PN junction, Depletion region, built-in potential, Working of solar cell,	7

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	Photo diode and LEDs.	
4	**Electromagnetic Theory:** Physical significance of gradient, divergence and curl, Concept of Displacement current, Maxwell's equations, Poynting Vector & theorem, EM waves and its propagation in free space and conducting medium, skin depth, Applications of EM waves in Wireless Communication (qualitative).	**7**
5	**Engineering Optics:** Laser: Basic of Laser and characteristics, Einstein's coefficient, Ruby laser, He-Ne laser and semiconductor laser, Applications of Laser- LIDAR, holography and optical information security. Optical fibre: Structure of an Optical fibre; Acceptance angle, Numerical aperture, V-Number, Classification of optical fibre, fibre Losses, Advantages and applications of optical fibre- Data transmission, medical imaging, fibre optic communication system.	**7**

Prescribed Text Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Physics Vol. 1 & 2	S. L. Gupta	5th	Shubham Publications	2016
2	Engineering Physics- I & II	S. K. Gupta	26th	Krishna Prakashan,	2023
3	Engineering Physics Vol I & II,	S. Prakash, A. Kumar, S.L. Rajput, C. Garg	1st	Pragati Prakashan	2015
4	Engineering Physics	Navneet Gupta	6th	Dhanpat Rai & Co.	2015
Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Concept of Modern Physics	A. Beiser	6th	McGraw Hill	2012
2	Introduction to Nanotechnology	C. P. Poole, Jr. and F. J. Owens	1st	Wiley Edition	2009
3	Solid State Electronic devices	B. G. Streetman and S. K. Banerjee	6th	Prentice Hall	2011
4	Physics of Semiconductor devices	S. M. Sze. And K.K. Ng	3rd	John Wily & Sons, New Delhi	2008
5	Introduction to Electrodynamics	D. J. Griffiths	2nd	Pearson Education India Learning Private Limited	1997
6	Optical Fibre	G. Keiser	5th	McGraw Hill	2015

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	Communication				
7	Lasers: Theory and Applications	K. Thyagarajan and A.K. Ghatak	1st	Macmilan India Ltd., New Delhi	2001
8	Introduction to Fourier Optics	J. W. Goodman	2nd	Roberts & Company	2005
Web References					
1. https://nptel.ac.in/courses/113106093					
2. https://nptel.ac.in/courses/115103039					

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25AS101/25AS201)

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	**Statement**	**Knowledge Level, KL**
CO1	Apply Schrödinger wave equation and matter wave concepts to solve problems related to particle confinement, quantum tunneling, and quantum computing.	**K3**
CO2	Analyze quantum confinement in nanomaterials and evaluate spintronics and Giant Magnetoresistance (GMR) for memory device applications.	**K4**
CO3	Analyze the operation and performance of semiconductor devices, including solar cells, photodiodes and LEDs, for engineering applications.	**K4**
CO4	Apply Maxwell's equations and electromagnetic wave propagation concepts to analyze wireless communication systems.	**K3**
CO5	Apply the principles of lasers and optical fibres to analyze the working and performance of optical communication, LIDAR and optical information security systems.	**K3**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2						2	2		3
CO2	3	2				3		2	2		3
CO3	3	2				3		2	2		3
CO4	3	2				3		2	2		3
CO5	3	2				3		2	2		3
Course Avg.	3.00	2.00				3.00		2.00	2.00		3.00

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25AS101/25AS201)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term - I	CO1(10 marks) +CO2 (5 marks)	15
Mid Term -II	CO2 (5 marks) +CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech.	I / II	Basics of Mechanical & Automation Engineering	25ME101 / 25ME201	2	1	0	3

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
36	12	0	42	90

Course Objective (s)
The course aims to build interdisciplinary awareness and practical insight into modern engineering applications. It provides all engineering students with a strong foundation in key concepts of Mechanical Engineering and Automation, applicable across domains such as design, manufacturing, thermal, energy systems, automation, and control engineering.

Syllabus

Unit No.	Topics	CI Hours
1	**Engineering Mechanics:** Force moment and couple, principle of transmissibility, Resultant of force system- concurrent Types of supports (Hinge, Roller) and loads (Point, UDL, UVL), free body diagram, equilibrium equations and Support Reactions. **Strength of Materials:** Normal and shear Stress, Strain, Hookes' law, Poisson's ratio, elastic constants and their relationship, stress-strain diagram for ductile and brittle materials, factor of safety.	9
2	**Thermodynamics:** Introduction to Thermodynamics, system and surrounding, basic thermodynamic terminologies (equilibrium, state, property, process, cycle), reversible and irreversible process, laws of thermodynamics (with numerical on first law of thermodynamics), comparison among heat engine and heat pump. **Refrigeration:** Refrigerating Effect, Ton of Refrigeration, Coefficient of Performance, Principle of Vapour Compression Refrigeration System, Construction, and Working of Domestic Refrigerator. **Air-Conditioning:** Introduction and Applications of Air Conditioning, Psychrometric Properties and Construction and Working of Window Air Conditioner.	9

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3	**IC Engine:** Definition and Basic Components of an Engine, Construction and Working of Two-Stroke and Four-Stroke Spark and Compression Ignition Engines, Comparison Between Two-Stroke and Four-Stroke SI and CI Engines, Types of fuels. **Electric vehicles and hybrid electric vehicles:** Working principle and components of an EV, advantages and disadvantages of EVs. working of hybrid electric vehicles, series and parallel types of HEV, degree of hybridization, advantages and disadvantages of HEV.	**6**
4	**Sensors and Transducers:** Introduction to Sensors, transducers and their applications, basics of Feedback and Control: Open-loop vs Closed-loop Systems. **Overview of Mechanical Actuation System:** Overview of kinematic chains, cam mechanisms, ratchet mechanisms, gears and their types, belts, and bearings. **Hydraulic and Pneumatic Actuation Systems:** Overview of fluid power systems, Pressure and direction control valves, rotary actuators, basic hydraulic and pneumatic sequencing problems	**6**
5	**Robotics:** Introduction and significance in automation, Classification (articulated, cartesian, cylindrical robots, etc.); Robot specifications (DOF, payload capacity, reach, repeatability), Introduction to SCARA robots. **3D Printing:** Introduction and types of 3D printing technologies, materials used (plastics, metals, composites), 3D printing process workflow, applications and limitations.	**6**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Engineering Mechanics	S.S. Bhavikati & K.G. Rajasekharappa	2nd	New Age	2010
2	Fundamentals of Mechanical Engineering	Dr. D. S. Kumar	1st	Katson Books	2023
3	Engineering Thermodynamics	R. K. Rajput	3rd	Laxmi Publications	2007
4	Refrigeration and Air Conditioning	R.S. Khurmi	1st	S. Chand	2006
5	Internal Combustion Engines	V. Ganesan	3rd	McGraw-Hill Education	2010
6	Hybrid Electric Vehicles Energy Management Strategies	Simona Onori,	2nd	London Springer	2022

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7	Robotics and Industrial Automation	R. K. Rajput	1st	S. Chand	2008
8	Additive Manufacturing Foundation Knowledge for the Beginners	Sunpreet Singh	1st	Springer	2021
Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Engineering Mechanics	S. S. Bhavikatti	7th	New Age International	2019
2	Strength of Materials	S.S. Rattan	2nd	McGraw-Hill	2013
3	Thermodynamics: An Engineering Approach	Yunus Cengel, M.A. Boles,	2nd	Tata McGraw Hill, New Delhi	2011
4	Refrigeration and Air Conditioning	C.P. Arora	4th	McGraw-Hill	2021
5	Electric and Hybrid Vehicles	Iqbal Husain & M. Ehsani	3rd	CRC Press / Springer	2021
6	Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering	W. Bolton,	4th	Pearson Education, New Delhi	2011
7	Introduction to Industrial Robotics	Ramachandran Nagarajan,	1st	Pearson India	2016
8	3D Printing for Dummies	Richard Horne	2nd	Wiley India	2017
Web References					
1. nptel.ac.in/courses/112106286					
2. nptel.ac.in/courses/112106141					
3. nptel.ac.in/courses/112108148					
4. nptel.ac.in/courses/112105128					

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25ME101/25ME201)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Apply the principles of engineering mechanics and analyze mechanical systems using concepts of force, stress-strain and material properties.	**K3**
CO2	Analyze thermodynamic systems and apply the laws of thermodynamics to basic energy systems, including HVAC applications.	**K3**
CO3	Explain the construction, working principles and comparative aspects of IC engines, electric vehicles, and hybrid vehicles.	**K2**
CO4	Describe the working principles and functions of sensors, actuators, and control elements used in basic automation and mechanical actuation systems	**K2**
CO5	Identify and interpret the applications of robotics and 3D printing in smart and automated industries.	**K2**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	3	1		1				2		2
CO2	3	3	1		1	2			2		3
CO3	1	2	1		2	2			2		3
CO4	1	1	1	1	2				2		3
CO5	1	1	1	1	2	2			2		3
Course Avg.	1.80	2.00	1.00	1.00	1.60	2.00			2.00		2.80

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN ((25ME101/25ME201)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term - I	CO1(10 marks) + CO2(5 marks)	15
Mid Term -II	CO2 (10 marks) + CO3 (5 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech	I/II	Basics of Electrical Engg.	25EL101/ 25EL201	2	1	0	3

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
36	12	0	42	90

Course Objective (s)
1. To make students understand the fundamentals of electrical circuits, voltage current calculations for various circuit elements connected in series-parallel configurations.
2. To make students understand the operating principle of different types of AC and DC machines and emerging trends in modern power system.

Syllabus

Unit No.	Topics	CI Hours
1	Introduction to electrical networks and classification of circuit elements: active and passive, linear and nonlinear, unilateral and bilateral, lumped and distributed, Types of sources and source transformation techniques, Application of Kirchhoff's laws (KVL and KCL), Mesh analysis and nodal analysis, Network theorems: Superposition theorem, Thevenin's theorem and Norton's theorem (with independent sources only)	7
2	Single-phase circuits: Analysis of single-phase R, L, C, RL, RC, and RLC series circuits, Complex notation-based analysis of parallel RLC circuits, Active, reactive, and apparent power, Resonance in AC circuits: Series and Parallel resonance and their resonance curves, Three-phase circuits: voltage and current relations in balanced star and delta connections	8
3	Magnetic circuits: Analogy between electric and magnetic circuits, Faraday's law of electromagnetic induction, B-H loop and hysteresis losses, Transformers: working principle, EMF equation, ideal and practical transformers, Equivalent circuit, losses, efficiency, and numerical problems on efficiency.	7
4	DC Machines: Construction, working principle, types, EMF equation and numerical related to EMF equation, torque equation of motor and applications, Three-phase Induction Motors: Operating principle, concept of slip and numerical problems related to slip, applications of three phase induction motor, Single-phase Induction Motors: working principle and	7

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	applications	
5	Overview of the electrical power system with single-line diagram, Introduction to grid systems, micro-grids, and smart grids, Distributed Generation (DG): Types, benefits, and challenges, Need and process of Earthing, Batteries: Classification, efficiency and backup calculations, Battery Management System (BMS): Block diagram and advantages, Battery Energy Storage System (BESS): Need and advantages in modern electrical networks, Role and applications of Artificial Intelligence (AI) in modern power systems.	**7**

Prescribed Text Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Basic Electrical Engineering	D P Kothari, I J. Nagrath	4	McGraw Hill	2019
2	Basic Electrical Engineering	J. B. Gupta	17	Katson	2025
3	A Textbook of Electrical Technology Vol 1 & II	B. L. Theraja, A. K. Theraja	23	S. Chand	2023
4	*Electrical Engineering: Concepts and Applications*	P. V. Prasad, S. Sivanagaraju	1	Cengage	2012
Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Electrical and Electronics Technology	Edward Hughes, John Hiley, Keith Brown, Ian McKenzie-Smith,	12	Pearson	2025
2	Electrical Engineering Fundamentals,	Vincent Del Toro	2	Pearson India Education	2015
3	Basic Electrical Engineering	D. C. Kulshreshtha	2	McGraw Hill	2019
4	Fundamentals of Electrical Engineering	L. S. Bobrow	2	Oxford University Press	1996

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Web References
1. https://www.nptel.ac.in/courses/108105112?utm
2. https://www.nptel.ac.in/courses/108105053?utm
3. https://share.google/vLdrZMI96v8t0yylq

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25EL101/25EL201)

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	**Statement**	**Knowledge Level, KL**
CO1	Apply Kirchhoff's laws and network theorems to analyze and solve direct current (DC) electrical circuits	**K3**
CO2	Determine the steady-state response, power, and resonance characteristics of AC circuits using circuit principles.	**K3**
CO3	Demonstrate an understanding of magnetic circuit principles and apply them to the operation and analysis of single-phase transformers.	**K3**
CO4	Demonstrate the construction, working principles, and applications of DC machines and induction motors for industrial applications.	**K3**
CO5	Analyze the role and integration of emerging technologies in modern electrical power systems.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2									
CO2	3	1									
CO3	3	2				3					
CO4	3	1				3					
CO5	3	1				3					3
Course Avg.	3	1.4				3					3

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25EL101/25EL201)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	**Assessment Tool**	**Remarks**	**Weightage of Marks in CIA**
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			**30**

Assessment Plan for External Component		
Assessment Tool	**COs Coverage**	**Weightage of Marks in External Component**
Mid Term - I	CO1(10 marks) +CO2 (5 marks)	15
Mid Term -II	CO2(5 marks) +CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		**70**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	I / II	Basics of Electronics Engineering	25EC101/ 25EC201	2	1	0	3

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
36	12	0	42	90

Course Objective (s)
1. Develop an understanding of semiconductor devices, integrated circuits, communication systems, and emerging technologies, including 5G/6G and IoT, to analyze their operating principles, characteristics, fabrication processes, and applications in modern electronic and smart systems.
2. Apply the fundamentals of digital electronics to analyze and design combinational logic circuits and understand the basics of digital signal processing, quantum computing, and edge computing.

Syllabus

Unit No.	Topics	CI Hours
1	**Unit I: Diode & its Applications** **PN junction diode:** Working Principle, V-I Characteristics, Rectifiers - Half Wave & Full Wave, PIV, Ripple Factor and Efficiency, Clippers & Clampers **Special Purpose Diodes:** Construction, Characteristics and Applications - Tunnel Diode, Schottky Diode, Varactor Diode, Photodiode, Zener Diode	8

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2	**Unit II: Transistor and its Applications** **BJT:** Construction & Working Principle of Bipolar Junction Transistor, Common Base, Common Emitter and Common Collector Configurations and their characteristics, BJT as a switch **FET:** Comparison of BJT & FET; Construction, Working Principle and Characteristics of JFET & MOSFET (Enhancement and Depletion), Concept of pinch-off, FET as a voltage variable resistor and a constant current source **CMOS:** Construction, Working Principle and Characteristics of CMOS, NOT, NAND & NOR implementation using CMOS	**7**
3	**Unit-III: Digital Electronics** Number Systems, Conversions, Codes (Weighted and Non-weighted codes, BCD & Gray Code), Boolean Algebra, Complements, Binary Arithmetic, Logic Gates, Boolean Expressions (SOP & POS forms), Minimization using K-Map (up to 4 variables), don't care condition and realization using logic gates Sampling & Quantization in Digital Signal Processing Fundamentals and Applications of Quantum Computing and Edge Computing	**8**
4	**Unit-IV: Basics of Integrated Circuits** **Introduction:** Evolution from discrete components to IC, Moore's Law and Scaling, Monolithic v/s Hybrid IC, Levels of Integration (SSI, MSI, LSI, VLSI and ULSI) **Overview of IC fabrication techniques:** Wafer preparation, Oxidation, Photolithography, Etching, Doping (Diffusion and Ion implantation), Deposition, Metallization and Packaging **Op-Amp:** Pin Configuration, Ideal Characteristics, Applications as Inverting Amplifiers, Non-Inverting Amplifiers **555 Timer**: Pin Configuration, Astable and Monostable operations, Applications as Waveform Generators and Delay Circuit.	**8**
5	**Unit-V: Communication Technologies** **Introduction:** Elements of Communication System, Modulation, Need of modulation **Cellular Networks**: An overview, Different generations in cellular communication systems, Introduction to 5G and 6G **Smart Technology:** Introduction to IoT Architecture, Sensors and Transducers, Wireless Sensor Network	**5**

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Prescribed Text Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Electronic Devices and Circuit Theory	Robert L. Boylestad / Louis Nashelsky	11th Edition	Pearson Education	2013
2	Digital Design	M. Morris Mano, Michael D. Cilleti	6th Edition	Pearson Education	2019
3	Electronic Communication Systems	George Kennedy and Bernard Davis	4th Edition	Tata McGraw Hill	2009
Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Electronic Devices and Circuits	Jacob Millman, C.C. Halkias, Satyabrata Jit	3rd Edition	McGraw Hill	2010
2	Fundamental of Digital Circuits	A. Anand Kumar	4th Edition	PHI	2018
3	Basic Electronics	D. P. Kothari and I. J. Nagrath	2nd Edition	McGraw Hill	2017
4	Digital Electronics	S Salivahanan and S Arivazhagan	1st Edition	Vikas Publications	2011
Web References					
1. https://nptel.ac.in/courses/122106025					
2. https://nptel.ac.in/courses/108101091					
3. https://www.coursera.org/learn/basic-electronics-odc					
4. https://www.youtube.com/watch?v=ZXOXAyouLOA					

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25EC101/25EC201)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Understand the working principle and characteristics of diodes and their various applications.	**K3**
CO2	Understand the working principle and characteristics of Transistors (BJT, FET & CMOS) in various configurations.	**K2**
CO3	Understand the concepts of number systems, logic gates, binary arithmetic, Boolean algebra and Karnaugh maps for digital circuit design.	**K3**
CO4	Apply Op-amp & IC-555 in various practical applications.	**K3**
CO5	Understand the fundamentals & recent trends in Communication Engineering.	**K2**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	2	1							2		
CO2	2	2	1	2	1				2		
CO3	3	3	2	3	3				2		3
CO4	3	3	3	3	3				2		1
CO5	2	1			2	3	3		2		3
Course Avg.	2.40	2.00	2.00	2.67	2.25	3.00	3.00		2.00		2.33

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25EC101/25EC201)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term - I	CO1(10 marks) + CO2 (5 marks)	15
Mid Term -II	CO2(5 marks) + CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	I	Fundamentals of Linear Algebra & Statistics	25AS102	2	1	0	3

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
42	12	0	36	90

Course Objective (s)
1. The objective of a linear algebra course for engineering students is to understand matrix algebra, vector spaces and linear transformations to analyze and solve problems related to linear systems, data analysis, and other engineering applications.

Syllabus

Unit No.	Topics	CI Hours
1	**Unit 1: Vector Spaces** Vector space, subspace, elementary properties of vector spaces, sum of subspaces, linear combination, linear span, linear dependence and independence, basis, dimension and range.	8
2	**Unit 2: Linear Transformation** Linear transformation, Rank, nullity, kernel and image of linear transformation. Rank-nullity theorem, matrix of linear transformation, change of basis, Inverse of a linear transformation, matrix associated with a linear map and its application in Engineering.	9
3	**Unit 3: Matrices and its Applications** Types of Matrices, Inverse of a matrix using Gauss Jordan Method, Rank of a matrix, Echelon form, Normal form of a Matrix, System of linear homogeneous and non-homogeneous equations, characteristic equation of a matrix, Eigen values, Eigenvectors and Cayley Hamilton theorem (without proof), Application of Matrices.	9

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4	**Unit 4**: **Statistical Techniques -I** Introduction, Collection and Classification of data, Graphical Representation, Measures of central tendency, Measures of dispersions (Range, Quartile deviation, Mean deviation, Standard deviation), Coefficient of variation, Standard deviations of combination of two groups, Moments, Skewness, Kurtosis.	**8**
5	**Unit 5: Statistical Techniques -II** Curve Fitting, Method of least squares, fitting of straight lines, Fitting of second-degree parabola, Exponential curves, Correlation and Karl Pearson's coefficients of correlation, Rank correlation, Regression (linear and non-linear), Multiple Linear Regression and their applications in engineering analysis, prediction, and decision-making.	**8**

Prescribed Text Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Outline of Linear Algebra	S. Lipschutz	4th	Schaum Series	2008
2	Statistical Methods	S. P. Gupta	46th	Sultan Chand & Sons	2023
3	Liner Algebra and its applications	David C. Lay, Steven R. Lay	6th	Pearson	2020
Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Linear Algebra	K. Hoffman and R. Kunze	2th	Prentice Hall of India	2015
2	An Introduction to Linear Algebra	V. Krishnamurthy, V. P. Mainra and J. L. Arora	1st	Affiliated East-West	2016
3	Linear Algebra	Stephen H. Friedberg	4th	PHI	2014
4	Mathematical Statistics with Applications.	Wackerly, Dennis D., Mendenhall III, William & Scheaffer, Richard L.	6th	Cengage Learning	2001

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5	Fundamentals of Mathematical Statistics	Gupta, S. C. and Kapoor, V.K.	4th	Sultan Chand & Sons	2008
Web References					
1. https://ocw.mit.edu/courses/8-05-quantum-physics-ii-fall-2013/04b0570b349e84d74129eef504498472_MIT8_05F13_Chap_03.pdf					
2. https://home.iitm.ac.in/anujjakhar/matrices.pdf					
3. https://www.cimt.org.uk/projects/mepres/alevel/stats_ch12.pdf					
4. https://youtu.be/fNk_zzaMoSs?si=iQRek1EKS6iENxx8					

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25AS102)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Understand the structure of vector spaces, subspaces, basis, dimension, and their properties.	**K2**
CO2	Apply the concept of linear transformations to solve the Engineering problems.	**K3**
CO3	Understand and apply the concepts of matrix algebra, systems of equations, eigen values and eigen vectors to solve engineering and scientific problems.	**K3**
CO4	Apply measures of central tendency and dispersion to analyze and interpret real-world data.	**K4**
CO5	Analyze data sets using correlation, regression, and curve fitting techniques.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	2	1	1		1				1		1
CO2	2	1	1		1				1		1
CO3	2	1	1	1	1				1		1
CO4	2	1	2	2	1				2		1
CO5	2	1	2	2	1				2		1
Course Avg.	2.00	1.00	1.40	1.66	1.00				1.40		1.00

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25AS102)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term – I	CO1(10 marks) +CO2 (5 marks)	15
Mid Term -II	CO2(5 marks) +CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech	I	Programming in C++	25CS101	3	0	0	3

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
36	0	0	54	90

Course Objective (s)
Understanding the basics of computers and programming.
Understanding and application of C++ as a programming language.
Solving basic string and array related problems.

Syllabus

Unit No.	Topics	CI Hours
1	Introduction to components of a computer system: Memory, processor, I/O devices, storage, and operating system. Concept of assembler, compiler, interpreter, loader and linker. Idea of program/algorithm: Representation of algorithms - flowchart, and algorithms. Visualization of flowchart and algorithms. From algorithms to programs, and source Code. Programming basics: Writing and executing the first C++ program, syntax and logical errors in compilation with examples. Object and executable code. The structure of a C++ program, the main function, header files, standard output/input, and comments**.**	6
2	Tokens in C++: Keywords, identifiers/variables, constants, punctuators, and operators. Variables: Variable naming convention, scope and their lifetime. Data types: Data types classification - basic, derived and user defined. Basic data types: Data types, their storage requirements, and typecasting. Operators: Arithmetic and miscellaneous operators, relational operators, logical operators, conditional operators, and bitwise operators. Associativity and precedence. Expressions and their evaluation.	8

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	Introduction and usage of Math library functions. Taking input on coding platforms - 1: Single value in a single line, multiple known values in a single line, multiple known values in multiple lines. Program flow control: if statement, if - else statement, nested if - else, else - if ladder, and switch - case.	
3	Iteration and loops: Use of while, do-while and for loops, multiple loop variables, use of break, and continue statements. Nested iteration. Taking input on coding platforms - 2: Single value in a single line, multiple known values in a single line, and others. Arrays: Array notation and representation, manipulating array elements, using multi dimensional arrays. Introduction to dynamically growable arrays (vectors in STL), Taking inputs on coding platforms - 3: Multiple unknown values in a single line, multiple unknown values in multiple lines.	**8**
4	Functions: Introduction, types of functions, passing parameters to functions, call by value, call by reference. Introduction to recursive functions.	**6**
5	Strings: String representation in C++. String manipulation operations with the help of functions from string.h library, specifically, replacing, inserting and erasing characters, finding character or substrings, counting characters or substrings, finding length of a string, case conversion, dealing with alphanumeric strings, concatenating two or more strings, and comparing two strings. Introduction to Object Oriented Programming: Discussion on characteristics of OOP, basic understanding of class and objects with examples, properties and methods in a class. Basic structure of a class in C++ and creating objects.	**8**

Prescribed Text Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	C++ Primer .	Stanley B. Lippman, Josée Lajoie, Barbara E. Moo	5th Edition	Addison-Wesley Professional	2012
2	Object-Oriented Programming with C++	E. Balagurusamy	8th Edition	McGraw Hill Education (India) Private Limited	2021
Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**

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1	C++: The Complete Reference	Herbert Schildt	4th Edition	McGraw-Hill Education (Osborne/McGraw-Hill; Tata McGraw-Hill in India)	2003
2	Object-Oriented Programming in C++	Robert Lafore	4th Edition	SPearson Education India / SAMS	2008

Web References
1. https://www.programiz.com/cpp-programming
2. https://www.geeksforgeeks.org/c-plus-plus
3. https://leetcode.com
4. https://www.codechef.com

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25CS101)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	To understand simple algorithms for arithmetic and logical problems.	**K3**
CO2	Demonstrate understanding of data types, operators, and control statements, and apply them in writing simple C++ programs.	**K3**
CO3	Understand arrays in C++ and apply iterative structures to perform operations on them in programs.	**K3**
CO4	Understand the modular approach to coding and solve problems by decomposing them into functions.	**K3**
CO5	Understand strings in C++, analyze them to identify patterns for solving string-related problems, and demonstrate knowledge of basic object-oriented concepts.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	1	1	2						
CO2	3	2	2	2	2						
CO3	3	2	3	2	3						
CO4	3	3	3	3	3						
CO5	3	3	3	2	3						
Course Avg.	3.00	2.40	2.40	2.00	2.60						

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25CS101)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term – I	CO1(10 marks) +CO2 (5 marks)	15
Mid Term -II	CO2(5 marks) +CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	I/II	Environment & Sustainability	25AS103/ 25AS203	2	0	0	2

Teaching & Learning Scheme (in hours per semester)

Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
24	0	0	36	60

Course Objective (s)

1. To introduce the basic concepts of sustainability practices and analyze the role of engineers in environmental management
2. To explore pollution control technologies, environmental monitoring, and smart resource management and evaluate alternative energy systems and digital innovations supporting sustainable development.
3. To integrate sustainability concepts into engineering design, operational practices and to familiarize with environmental laws, policies, ethics, and compliance requirements

Syllabus

Unit No.	Topics	CI Hours
1	Sustainable Development: Pillars, Goals (17 SDGs) and Challenges, Framework to assess the impacts of anthropogenic activities, Role of Engineers in Sustainable Development	**6**
2	Water Treatment Technologies and Smart Solution, Air Quality Index and air pollution control devices, Carbon footprint, Earth overshoot day, Resource depletion and management, Environmental Resilience, Blockchain: Sustainability concerns, Integrated Solid Waste Management, Urban mining and Bioremediation	**6**
3	Fossil Fuel Dependence and Decentralized Energy Systems, Hydrogen as a future fuel and its environmental implications, Biofuels and their Ecological Trade-offs, Electric vehicles (EVs): environmental impact, AI/ML in environmental prediction, Smart and Sustainable agriculture, Green computing	**4**
4	Sustainable Practices: ISO 14001, Material Life Cycle Assessment, Ecological Footprint and Clean Development Mechanism (CDM), Green Buildings, Bureau of Energy Efficiency (BEE) Star Label, Environmental Degradation and Probable Solution of some Emergent Technologies- Additive Manufacturing (3D printing), Nanotechnology, Electromagnetic Radiations	**4**

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5	International Environmental Agreements: Conventions, Protocols and Treaties, The Environment (Protection) Act, 1986 (EPA,1986), Environmental Impact Assessment (EIA) in engineering projects, Polluter Pay Principle (PPP)/Extended Producer Responsibility (EPR).	**4**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Sustainable Engineering: Principles and Practice	Bakshi, B.R.	1st	Cambridge University Press	June 2019
2	Introduction to Environmental Engineering and Science	Hamza et al.	1st	Bright Sky Publications	2024
3	Textbook of Environment and Ecology	Dave, Katewa & Singh	2nd	Cengage Learning India Pvt Ltd Delhi	2025

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Introduction to Environmental Engineering and Science	Masters, G.M. and Ela, W.P.	3rd	Pearson Education	2015
2	Sustainable Engineering: Concepts, Design and Case Studies	Allen, D. T. and Shonnard, D. R		Prentice Hall	
3	Basic Concepts in Environmental Management	Mackenthun, K.M	1st	CRC Press	1999

Web References

1. nptel.ac.in/courses/103107215
2. Documentary The Bhopal Disaster INDIA Nat Geo Full 2014 720p HD

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25AS102/25AS202)

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	**Statement**	**Knowledge Level, KL**
CO1	Implement sustainability frameworks, SDGs, and the engineering role in mitigating anthropogenic impacts	**K3**
CO2	Impart knowledge on pollution abatement technologies and environmental monitoring techniques.	**K3**
CO3	Understand renewable energy options and recognise career paths in the green sector.	**K2**
CO4	Apply sustainability standards, life cycle assessment, and green building frameworks in engineering design	**K3**
CO5	Interpret key environmental laws, global treaties and environmental compliance standards in professional contexts	**K2**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	1		2	3	3	2		2	3
CO2	3	3		1	2	2	3	1			3
CO3	2	3	1	1	2	2	2	1			3
CO4	2	3	2	2	2	2	2	1		2	3
CO5		1	1	1	2	3	3	2			3
Course Avg.	2.50	2.40	1.25	1.25	2.00	2.40	2.60	1.40		2.00	3.00

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25AS102/25AS202)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term - I	CO1(10 marks) +CO2 (5 marks)	15
Mid Term -II	CO2(5 marks) +CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech	I/II	Essentials of Artificial Intelligence	25AI101/ 25AI201	2	0	0	2

Teaching & Learning Scheme (in hours per semester)

Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
24	0	0	36	60

Course Objective (s)

1. The objective of this course is to provide students with a foundational understanding of Artificial Intelligence (AI), its core concepts, techniques, and real-world applications.
2. The course aims to introduce fundamental principles of intelligent systems, problem-solving, machine learning, and reasoning while fostering critical thinking about the opportunities, challenges, and ethical considerations of AI.
3. By the end of the course, students will be equipped with the essential knowledge and practical skills to understand, evaluate, and apply AI methods.

Syllabus

Unit No.	Topics	CI Hours
1	Introduction to Artificial Intelligence (AI): Need, what and what is not AI, Human Intelligence v/s Artificial intelligence, Types of AI, Real life application, Relationship among AI, Machine Learning (ML) and Deep Learning (DL). How does AI work? History, Future in AI, Recent Trends in AI, Career Trajectories with AI, How AI affect jobs? Statistical Foundations: Types of data, Descriptive Statistics (Measures of Central Tendency, Measures of Dispersion), Probability and distribution (Probability Rules, Normal/Binomial/Poisson Distributions)	5
2	Definition, Types of Agents, Characteristics, Agent Communication, Agent Negotiation, PEAS Description, Typical Intelligent Agents, Introduction to Problem Solving Approach to typical AI problems, Knowlege Representation basics -Predicate logic and Preposition logic.	5
3	Introduction to Python, Datatypes, Operator, control statements and loops, Function, Python Package (Numpy, Pandas, Matplotlib, scikit-learn), importing of datasets - file format, Data Preprocessing.	5

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4	AI Interfaces: Kaggle, Google Colab, Jupyter, Case studies with data on Kaggle, Kaggle Competitions. Prompt Engineering: Need, Art of Prompting, Prompt for AI, Prompting Techniques (zero-shot prompting, few-shot prompting, chain-of-thought prompting), Types, Use cases, Benefits of Prompt Engineering, Future of prompt engineering, Career and role in prompt engineering.	**5**
5	Modern AI Approaches & Ethics: Discriminative vs Generative AI, Edge AI, XAI, Agentic AI, Applied AI, Large Language Models, AI Tools: Microsoft Co-Pilot, DALL-E, ChatGPT, Ethics in AI- Safety and Security, Ethical and legal challenges, Trust and adaptation, Case Studies, Future risk.	**4**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Artificial Intelligence: A Modern Approach	Stuart Russell and Peter Norvig	4th edition	Pearson	2021
2	Python Machine Learning	Sebastian Raschka and Vahid Mirjalili	3rd Edition	Packt Publishing	2019
3	The AI-Powered Entrepreneur	Ethan Mollick	Ist Edition	Harvard Business Review Press	2024

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Life 3.0: Being Human in the Age of Artificial Intelligence	Max Tegmark	Ist Edition	Alfred A. Knopf	2017
2	Deep Learning by	Ian Goodfellow, Yoshua Bengio, Aaron Courville,	Ist Edition	MIT Press	2016
3	Artificial Intelligence: A Guide for Thinking Humans	Melanie Mitchell, Farrar, Straus and Giroux	Ist Edition	FSG	2019
4	LLM Prompt Engineering for Developers,	John Berryman and Albert Ziegler	Ist Edition	O'Reilly Media	2024

Web References

1. https://docs.python.org/3/
2. https://www.tensorflow.org/
3. https://huggingface.co/learn
4. https://ai.google/

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25AI101/25AI201)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Understand the fundamentals of AI, its applications, relation to ML/DL, and apply basic statistical concepts for data analysis.	**K3**
CO2	Understand intelligent agents and apply basic logic-based knowledge representation for AI problem-solving.	**K3**
CO3	Apply Python programming concepts and libraries to perform data handling, preprocessing, and visualization for AI/ML tasks	**K4**
CO4	Use AI development interfaces and analyze prompt engineering techniques.	**K3**
CO5	Analyze modern AI approaches and tools, and evaluate ethical, legal, and societal implications concerning safety, trust, and future risks in AI deployment.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	1				2					
CO2	3	1	1			3					
CO3	2	1	1	1	1						
CO4	1	1			2	3				2	
CO5	1			1	1	3	3				1
Course Avg.	2.00	1.00	1.00	1.00	1.33	2.75	3.00			2.00	1.00

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25AI101/25AI201)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term - I	CO1(10 marks) +CO2 (5 marks)	15
Mid Term -II	CO2(5 marks) +CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	I/II	Design Thinking, Innovation and Entrepreneurship	25HU102/ 25HU202	2	0	0	2

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
24	0	0	36	60

Course Objective (s)
1. Apply creative thinking techniques to generate, evaluate, and select innovative solutions for complex problems.
2. Design and develop prototypes of potential solutions and evaluate them through user testing and iterative improvement.
3. Foster teamwork, communication, and entrepreneurial thinking to create user-centric, feasible, and sustainable solutions with societal and business value.
4. Develop entrepreneurial competencies by identifying business opportunities, understanding value propositions, intellectual property rights (IPR), business models, and startup fundamentals.

Syllabus

Unit No.	Topics	CI Hours
1	Design Thinking Introduction Introduction to Design Thinking, conventional problem solving versus design thinking, business uses of design thinking, variety within the design thinking discipline, formation and importance of the design thinking team, workshops and meetings. Case Study: Mumbai Dabbawallas, Tata Nano, Jeevan Bindi	4
2	Design thinking approaches in stages: empathize with user, Moccasin Walk, define the problem, Ideate, Prototype (alternate solution), test the solution. Problem identification techniques and research methods Case Study: Portable Ultrasound Machine (GE V scan), Embrace Infant Warmer, Tally Software, Lucky Iron Fish Activity: Presentation (Case study)	4
3	Listening and empathizing techniques-engagement, observation, interviews, surveys, Define- Create problem statements, identify user needs and pain points, synthesize research data, Ideate- Pattern recognition, scamper techniques, and	6

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	connecting the dots, Prototype- Paper prototypes, Wireframes, Models or role-play scenarios, Forms of testing in design thinking Case study: Pure it, Royal Enfield, Big Basket, IDEO Water Pump, Airbnb, Oral-B toothbrush. Activity: Empathy map, Journey map	
4	Concept of innovation, innovation and creativity, difference between invention and innovation, types of innovation, product vs process innovation, value creation, social and grassroots innovation, benefits of innovation, Technology Readiness Level (measuring innovation)	**5**
5	Introduction to IPR, Types of IPR (copyright, patent, trademark, GI tag), advantages and disadvantages of IPR, prior art research techniques using tools such as Google Patent and other. Entrepreneurship concept, myths versus reality of entrepreneurship, entrepreneurship competencies, sensing business opportunity, Business model canvas and value proposition, start-up success and failure story, idea pitching (pitch deck development)	**5**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	A textbook on Design Thinking Principles, Processes and Applications	R Srinivasan Mohammed Ismail K Arulmozhi	First Edition 2025	S Chand	2025
2	The Design of Everyday	Things by Don Norman	Revised and Expanded Edition, 2013	Publisher: Basic Books	2013
3	Design Your Thinking: The Mindsets, Toolsets and Skill Sets for Creative Problem-Solving	Pavan Soni	November 2023	Penguin Random House India	2020
4	Design Thinking: A Beginner's Perspective	E. Balagurusamy	Revised Edition 2025	Publisher: McGraw Hill Education India	2024

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Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Change by Design: How Design Thinking Creates New Alternatives for Business and Society	Tim Brown,	First Edition 2009	Harper Business	**2009**
2	Creative Confidence: Unleashing the Creative Potential Within Us All	Tom Kelley & David Kelley	Revised Edition 2013	: Crown Business	2013
3	Design Thinking: A Framework for Applying Design Thinking in Problem Solving	Anuja Agarwal	Second Edition 2024	Cengage India	2024
4	Design Thinking for Dummies	Christian Mueller Roterberg	Reprint 2023	Wiley India Pvt. Ltd.	2023

Web References
1. https://nptel.ac.in/courses/110106124
2. mumbai dabbawala case study by harvard university - Google Search
3. lucky iron fish case study - Google Search
4. https://www.youtube.com/watch?v=LHe-K0VYqhU

Certification for the self-learning component should preferably be included in the assessment.

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Estd. 2000

COURSE OUTCOMES (25HU101/25HU201)

Course Outcomes (COs):
Upon completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Differentiate conventional problem-solving from design thinking through case studies and team dynamics.	**K2**
CO2	Apply the stages of the design thinking framework to develop user-centred solutions.	**K3**
CO3	Apply empathy, ideation, prototyping, and testing techniques to generate and refine solutions.	**K3**
CO4	Apply the concept of innovation, types of innovation, assess value creation, and evaluate readiness using TRL.	**K3**
CO5	Apply IPR concepts and entrepreneurship tools to design a business model and pitch an idea.	**K3**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_3
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1				1				3			
CO2		3	2	3				2	2		
CO3	2	3	3	3	3			2	1		
CO4		1	3	3	1						
CO5			3			3	3		2		1
Course Avg.	2.00	2.33	2.75	2.50	2.00	3.00	3.00	2.33	1.67		1.00

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25HU101/25HU201)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term - I	CO1(10 marks) +CO2 (5 marks)	15
Mid Term -II	CO2(5 marks) +CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B Tech	II	Soft Skills	25HU101/ 25HU201	2	0	0	2

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
24	0	0	36	60

Course Objective (s)
1. Students will be able to create substantial base by the formation of strong professional vocabulary for its application at different platforms and through numerous modes as comprehension, reading and writing.
2. Students will be able to use correct English in professional settings in speech and writing confidently.
3. Students will be able to identify and comprehend the main idea, and supporting details in audio/video clip, and paragraph.

Syllabus

Unit No.	Topics	CI Hours
1	Unit 1: Vocabulary building: Word formation, Subject verb agreement (Discussion and Quiz) 1. Word formation: Root word, Bound words, Affixes, Compound Words, Acronyms, Abbreviations, Conversion, Back formation, Blending, Clipping, Derivation, etc. 2. Vocabulary Building: Antonyms, Synonyms, Homophones 3. Subject Verb Agreement: Finite/ Non finite verb, Gerund, Infinitive, Participle. 4. Quiz	5
2	Unit 2: Listening Skills: Discussion of topic, Ted talk listening video to be played in the class (Summarizing the listening clip by group and Quiz) 1. Listening Process, Types, Barriers, Traits of a good listener 2. Ted talk video Play in the class	4

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	3. Summary discussion by group leaders in the class 4. Quiz	
3	Unit 3: Speaking Skills: Speaking mode, Non verbal and Para linguistic Cues, and Overcoming Stage fright (Speaking aloud with control on Body language and para linguistics; Option: Podcast/ Speech/Dialogue delivery and Quiz) 1. Speaking Modes Discussion: Extempore, Impromptu, Reading and Memorizing & Drill 2. Nonverbal Features Discussion & Drill 3. Paralinguistic Features Discussion & Drill 4. Quiz	**5**
4	Unit 4: Reading Skills: Discussion of the Topic, Reading Comprehension styles (Solving Reading passage or case study and Quiz) 1. Reading Style Discussion: Skimming, Scanning, Churning, & Assimilation 2. Discussion on the ways to develop good reading practices 3. Solving Case study based on Communication Barrier and Discussion Solving Reading comprehension 4. Quiz	**4**
5	Unit 5: Writing Skills: Incorporating 7Cs in writing, Situation based blog writing, Content organization for Leave, Complaint, Permission, and Internship letter writing (Discussion and Quiz) 1. Discussion on 7Cs with examples: Clarity, Conciseness, Courtesy, Correctness, Concreteness, Coherence, Completeness 2. Content organization in Introduction, Development of Idea, and Conclusion 3. Situation based Responding: Blog Writing, Publishing and Link Sharing 4. Drafting Content for letters: Leave, Complaint, Permission, and Internship letter/E-Mail writing with Block Format 5. Quiz	**6**

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Prescribed Text Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	English Grammar & Composition	Wren & Martin	Regular Edition 2026	S. Chand & Co Ltd New Delhi	2026.
2	Soft Skills (The Employability Skills for 21st Century)	Ritu Soryan	Regular	Katson Books New Delhi	2025.
3	Soft Skills for Everyone.	Jeff Butterfield	II	Cengage Learning India Pvt. Ltd; New Delhi	2020
4	Soft Skills	K Alex	IV	S. Chand & Co Ltd New Delhi	2025
Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Personality Development for Life and Work	Harold R. Wallace et. Al.	10th	Cengage Learning India Pvt. Ltd; New Delhi	2012.
2	Personality Development & Soft Skills	Barun K. Mitra	3rd	Oxford University Press, New Delhi	2024
3	Public Speaking	William S. Pfeiffer	Regular	Pearson, Delhi	2012
4	English Grammar & Usage	R.P.Sinha	Regular	Oxford University Press, New Delhi	2005

Web References

1. https://www.englishbix.com/engineering-vocabularywords/
   https://www.powerthesaurus.org/engineering/antonyms
   https://www.powerthesaurus.org/engineering/synonyms
2. https://www.leonardoenglish.com/blog/the-best-activities-to-improve-your-english-with-ted-talks
   https://sg.indeed.com/career-advice/career-development/listening-skills
   https://www.planetspark.in/communication-skills/listening-skills

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3. https://fscj.pressbooks.pub/publicspeaking/chapter/four-types-of-speeches/ https://www.scribd.com/presentation/231143671/Modes-of-Presentation https://www.studocu.com/in/document/aligarh-muslim-university/communication-skills/modesof-delivery-lecturenotes-on-communicative-skills/72864104
4. https://expertprogrammanagement.com/2018/04/7-cs-communication/ https://www.calilio.com/blogs/the-7-cs-of-communication https://www.collegenp.com/article/10-ways-students-can-improve-reading-comprehension-skill
5. https://managetrix.com/resources/how-to-write-leave-letter/ https://applicationfox.com/leave-application-for-student/ https://apttones.com/sample-letters-of-request-for-internship/#google_vignett

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25HU101/25HU201)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Prepare the basic level of proficiency in English language through correct use of grammar and vocabulary.	**K3**
CO2	Converse well with effective listening and speaking skills in English language.	**K3**
CO3	Understand and Utilize the appropriate non-verbal and para linguistic cues.	**K3**
CO4	Comprehend the reading extracts, case, reports, etc. effectively.	**K2**
CO5	Acquire the skills to write the simple and correct English for blog, and letters.	**K3**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1							3		3		2
CO2							2	2	3		2
CO3							2	1	3		2
CO4		1		1		2	3	1	3		2
CO5				1			3	1	3		2
Course Avg.		1		1		2	2.60	1.25	3.00		2

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25HU101/25HU201)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Internship Letter/ Blog	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term - I	CO1(10 marks) +CO2 (5 marks)	15
Mid Term -II	CO2(5 marks) +CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	I/II	Physics Lab	25AS151/ 25AS251	0	0	2	1

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
0	0	24	6	30

Course Objective (s)
The course aims:
1. To understand and experimentally verify the principles of diffraction and polarization of light, and analyze their applications in optical systems.
2. To investigate the fundamental concepts of electromagnetism and study their role in engineering and technological applications.
3. To understand the basic principles of quantum theory and experimentally explore quantum phenomena relevant to modern physics.
4. To study the electrical characteristics and behaviour of semiconductor devices through laboratory experiments and interpret their practical significance.
5. To understand the principles of optical fibre communication and apply experimental techniques to analyse signal transmission and related engineering applications.

Syllabus

Unit No.	Topics	CI Hours
1	To determine the energy band gap of a given semiconductor using p-n junction diode method.	2
2	To Study of V-I characteristics of a diode.	2
3	To determine E.C.E. of copper using tangent galvanometer.	2
4	To verify Stefan's law by electrical method.	2
5	Measurement of wavelength of a laser light by using single slit diffraction.	2
6	To calibrate the given ammeter and voltmeter by potentiometer.	2
7	To study the V-I characteristics of a solar cell.	2
8	Determination of Numerical Aperture of an optical fibre.	2

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9	To determine the bending losses of optical fibre.	**2**
10	To study the characteristics of a photodiode.	**2**
11	To determine Planck's constant using photo cell experiment.	**2**
12	To determine the specific rotation of cane sugar solution using Polarimeter.	**2**
13	To determine the wavelength of monochromatic light by Newton's Rings.	**2**
14	To determine the wavelength of spectral lines of mercury vapour lamp using plane transmission grating.	**2**
15	To study the variation of magnetic field along the axis of a current carrying coil and then to estimate the radius of the coil.	**2**
	**Any 10 experiments will be chosen from above list*	
Value Added Experiments		
1	Demonstration of Fourier optics applications.	**2**
2	To study the effect of temperature on solar cell efficiency.	**2**

Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Engineering Physics Practical	S.K. Gupta	2001	Krishna	2001
2	Engineering Practical Physics	Gupta &Kumar	2013	Pragati Edition	2013
Web References					
1. https://vlab.amrita.edu/					

****Certification for the self-learning component should preferably be included in the assessment.***

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COURSE OUTCOMES (25AS151/25AS251)

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	**Statement**	**Knowledge Level, KL**
CO1	Determine the various optical parameters by using optics and modern Physics principles behind the experiments.	**K3**
CO2	Compute the electrical and electronic properties of materials through experiments.	**K3**
CO3	Demonstrate the fundamental physical laws using electrical setups.	**K3**
CO4	Experimentally determine and calculate key physical properties of materials.	**K3**
CO5	Evaluate equipment performance and parameters using graphical and computational techniques.	**K3**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**
CO1	1	1		1		2		2			2
CO2	1	1		1		2		2			2
CO3	2	1		1				2			1
CO4	1	1		1				2			1
CO5	1	1		1				2			1
Course Avg.	1.20	1.00		1.00		2.00		2.00			1.40

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25AS151/25AS251)

Evaluation Pattern			
Internal Component (CIA)		**External Component**	**Total Marks**
PA1	**PA2**		
20 +5 (Attendance)	20 +5 (Attendance)	50	**100**

Assessment Plan for CIA				
	CO	**Assessment Tool**	**Remarks**	**Weightage of Marks in CIA**
PA1	**CO1,CO2,CO3,CO4,CO5**	Conduction of experiment (5), quiz (5), viva&lab (10)	25%, 25%, 50%	20
PA2	**CO1,CO2,CO3,CO4,CO5**	Conduction of experiment (5), quiz (5), viva&lab(10)	25%, 25%, 50%	20
	Attendance			10 Marks
Total Marks				**50**

Assessment Plan for External Component		
Assessment Tool	**COs Coverage**	**Weightage of Marks in External Component**
Experiment Conduction	CO1, CO2, CO3,CO4,CO5	25
Viva-Voce	CO1, CO2, CO3, CO4, CO5	25
Total Marks		**50**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech	I/II	Professional Communication Lab	25HU151/ 25HU251	0	0	2	1

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
0	0	24	6	30

Course Objective (s)
The course aims:
1. To develop effective verbal, non-verbal, and written communication skills for academic, professional, and workplace interactions.
2. To enhance students' presentation, public speaking, group discussion, and interpersonal communication skills through experiential learning.
3. To foster critical thinking, analytical reasoning, and professional communication for effective problem-solving and decision-making.
4. To cultivate teamwork, leadership, ethical communication, and professional etiquette required for collaborative work environments.
5. To strengthen employability by developing confidence, interview skills and technical communication.

Syllabus

Unit No.	Topics	CI Hours
1	**Self- Introduction** – To demonstrate the ability to introduce oneself and others effectively in academic and professional contexts.	4
2	**Group Discussion** - To participate in group activities and using appropriate verbal and non-verbal communication in team.	4
3	**Extempore** - To empower students to convey ideas with clarity and build confidence	4
4	**Role Play** - To enhance conversational skills and emotional learnings for workplace.	4
5	**Mock Interview** - To develop interview skills and build rapport with the interviewer.	4

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6	**Theme Presentation** – To practice the analytical & critical thinking, and argumentative skills.	**4**
	Other Activities – Value Addition Activities Open Parliament Speech Relay Spin the Yarn Software Based Activity Formal Debates Creative Writing	**6**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Professional Communication	R. Gupta 1st	1st	University Science Press (Laxmi Publications)2021	2021
2	Technical Communication: Principles and Practice	Meenakshi Raman & Sangeeta Sharma	3rd	Oxford University Press	2018
3	Communication Skills	Sanjay Kumar & Pushp Lata	2nd	Oxford University Press	2018
4	Professional Communication	Ronald B. Adler, Jeanne Marquardt Elmhorst, Kristen G. Lucas	3rd	McGraw-Hill Education	2019

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Soft Skills	S. Boyer	1st	SAGE Publications	2024
2	Business Communication Today	Courtland L. Bovee & John V. Thill	15th	Pearson	2024
3	How to Win Friends and Influence People	Dale Carnegie	Revised Edition	Simon & Schuster	2022
4	The Definitive Book of Body Language	Allan Pease & Barbara Pease	Revised Edition	Bantam Books	2017

Web References

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1. The Muse – Interview Preparation- https://www.themuse.com
2. https://www.freshersworld.com- Freshersworld – GD Preparation
3. Toastmasters International – Public speaking, leadership, and communication practice.
4. IEEE Professional Communication Society – Resources on technical and professional communication for engineers.
5. IEEE Learning Network – Professional Communication Courses – Technical writing and communication courses for engineering professionals.
6. https://www.linkedin.com/learning- LinkedIn Learning – Interview Preparation
7. AICTE – All India Council for Technical Education – Student learning resources and employability initiatives.

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25HU151/25HU251)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Demonstrate the ability to introduce one self and others effectively in academic and professional contexts.	**K3**
CO2	Engage in structured and formal conversations using appropriate verbal and non-verbal communication in team.	**K2**
CO3	Cultivate relevant technical style of communication and presentation at their work place and also for academic uses.	**K2**
CO4	Practice the analytical & critical thinking, and argumentative skills.	**K3**
CO5	Develop the language competency for enhancing interview skills	**K3**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1							2		3		1
CO2							3	3	3		2
CO3							3		3		3
CO4							3	3	3		2
CO5							2		3		3
Course Avg.							2.60	1.20	3.00	0.00	2.20

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25HU151/25HU251)

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	**100**

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	**CO 1, CO2**	Quiz Viva Conduction of Experiments	25%+25%+50%	20 Marks
PA2	**CO3, CO4, CO5**	Quiz Viva Conduction of Experiments	25%+25%+50%	20 Marks
	Attendance			10 Marks
Total Marks				**50**

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Experiment Conduction	CO1, CO2, CO3,CO4,CO5	25
Viva-Voce	CO1, CO2, CO3, CO4, CO5	25
Total Marks		**50**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	I/II	Basic Electrical Engineering Lab	25EL151/ 25EL251	0	0	2	1

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
0	0	24	6	30

Course Objective (s)
The course aims:
1. To familiarize students with basic electrical instruments and their safe operation in laboratory experiments.
2. To develop the ability to measure electrical quantities such as voltage, current, power, resistance, and energy using standard measuring instruments.
3. To verify fundamental electrical laws and theorems, such as Ohm's Law, Kirchhoff's Laws, Thevenin's Theorem, Superposition theorem through practical experiments.
4. To study the characteristics and operation of electrical machines**,** including transformers, DC machines, and AC machines.
5.To understand the behavior of AC and DC circuits and analyze their performance under different loading condition
6.To enhance analytical, observation, and report-writing skills through experimental analysis and interpretation of results.
7. To develop practical skills in wiring, circuit assembly, testing, and troubleshooting using standard laboratory equipment.
8. To promote safe laboratory practices and adherence to electrical safety standards during experimentation.

Syllabus

Unit No.	Topics	CI Hours
	Part-A (Conduct any 7 experiments)	
1	Verification of Kirchhoff’s laws	**2**
2	Verification of Superposition Theorem	**2**
3	Verification of Thevenin’s Theorem	**2**

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4	Connection and measurement of power consumption of a fluorescent lamp (tube light).	**2**
5	Determination of parameters of ac single - phase series RLC circuit.	**2**
6	Study of phenomenon of resonance in RLC series circuit and obtain resonant frequency.	**3**
7	Measurement of power and power factor in a single phase ac series inductive circuit and study improvement of power factor using capacitor	**2**
8	To perform polarity and voltage ratio test on a single phase transformer and determine its transformation ratio	**2**
9	Determination of efficiency of a single -phase transformer by performing load test.	**2**
10	To study running and speed reversal of a three phase induction motor and record speed in both directions	**2**
11	Determination of efficiency of a dc shunt motor by load test	**2**
12	Demonstration of cut-out sections of machines: dc machine, three phase induction motor and single-phase induction motor.	**3**
	Part-B (Conduct any 3 experiments)	
13	To study the working and control of two lamps in series and in parallel.	**2**
14	To measure energy consumption of a house-hold by using single phase energy meter.	**2**
15	To study and wiring of distribution board including power plug using isolator and MCB.	**2**
16	To simulate and study the charging and discharging characteristics of a battery and observe its voltage, current and state of charge (SoC) over time using Matlab.	**3**
17	To simulate a Battery connected to a DC load and study its behaviour under different loading condition using Matlab.	**3**

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Prescribed Text Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Basic Electrical Engineering	J.B.Gupta	16th	S.K.Kataria & Sons	2019
2	A Textbook of Electrical Technology	B.L Theraja and A.K.Theraja	24th	S.Chand	2023
Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	A Textbook of Electrical Technology	B.L Theraja and A.K.Theraja	24th	S.CHAND	2023
Web References					
1	https://www.electricaltechnology.org/2012/11/single-phase-energy-meter.html				
2	https://www.mathworks.com/help/simscape-battery/?utm				
3	https://www.electricaleasy.com/2014/04/polarity-test-of-transformer.html?utm				
4	https://www.electrical4u.com/kirchhoffs-laws/?utm				
5	https://www.allaboutcircuits.com/textbook/alternating-current/chpt-6/resonance-series-parallel-circuits/?utm				

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25EL151/25EL251)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Apply fundamental electrical circuit laws and theorems to analyze and verify linear DC electrical circuits through laboratory experiments	**K3**
CO2	Apply electrical wiring and measurement techniques to construct and test basic domestic electrical installations and lighting circuits.	**K3**
CO3	Perform standard tests on single-phase transformers, DC machines, and three-phase induction motors to determine their operating characteristics and efficiency.	**K3**
CO4	Analyze the performance of single-phase AC circuits by determining RLC parameters, resonance characteristics, power, and power factor through experimental investigations.	**K4**
CO5	Analyze the operating characteristics and performance of electrical machines and transformers based on experimental observations and test results.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	2	2		1				3	3		
CO2	2	2	2	1	3	3		3	3		3
CO3	2	1		1				3	3		
CO4	2	2		2				3	3		
CO5	2	1		1				3	3		
Course Avg.	2	1.60	2	1.20	3	3		3	3		3

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25EL151/25EL251)

Evaluation Pattern			
Internal Component (CIA)		**External Component**	**Total Marks**
PA1	**PA2**		
20 +5 (Attendance)	20 +5 (Attendance)	50	**100**

Assessment Plan for CIA				
	CO	**Assessment Tool**	**Remarks**	**Weightage of Marks in CIA**
PA1	**CO1**	Conduction Quiz Lab Files Assessment	25%+25%+50%	20
	CO2			
PA2	**CO3**	Conduction Quiz Lab Files Assessment	25%+25%+50%	20
	CO4			
	CO5			
	Attendance			10 Marks
Total Marks				**50**

Assessment Plan for External Component		
Assessment Tool	**COs Coverage**	**Weightage of Marks in External Component**
Experiment Conduction	CO1, CO2, CO3,CO4,CO5	25
Viva-Voce	CO1, CO2, CO3, CO4, CO5	25
Total Marks		**50**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	I/II	Electronics Engg. Workshop	25EC151/ 25EC251	0	0	2	1

Teaching & Learning Scheme (in hours per semester)				
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
0	0	24	6	30

Course Objective (s)
The course aims:
1. To provide practical knowledge of basic electronic components and their applications.
2. To develop an understanding of diode-based circuits and their operation.
3. To gain practical knowledge of IC-based circuits.
4. To develop practical knowledge of logic gates and digital circuits.
5. To apply Arduino in designing solutions for real-world engineering applications.

Syllabus

Unit No.	Topics	CI Hours
1	To study and identify the various electronic components.	2
2	Demonstration of bread board, CRO, DSO & DMM, and to study & practice of Soldering & desoldering.	2
3	To study the V-I of P-N junction diode and Zener diode.	2
4	To study the application of Zener diode as a voltage regulator and implementation of a regulated DC power supply using Breadboard.	2
5	To study the Logisim and realize the NOT, NAND & NOR gates using CMOS.	2
6	To study the transistor as a switch.	2
7	To realize the basic gates (AND, OR & NOT) using Universal logic gates (NAND & NOR).	2

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8	To implement the Boolean functions (SOP & POS) using basic gates.	**2**
9	To realize adder & subtractor circuit using OP-AMP (IC741).	**2**
10	To design an Inverting & non-inverting circuit having voltage gain 30.	**2**
11	To design a circuit which converts sine wave to square wave using 741C and IC 555, separately.	**2**
12	To study the basics of microcontroller & sensors.	**2**
	**Any 10 experiments will be chosen from above list*	
Value Added Experiments		
1	To realize a Temperature sensor display & Blinking of LED using Arduino.	**2**
2	To make a circuit-based project and prepare its report.	**2**

Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Electronic Devices and Circuit Theory	Robert L. Boylestad / Louis Nashelsky	11th Edition	Pearson Education	2013
2	Digital Design	M. Morris Mano, Michael D. Cilleti	6th Edition	Pearson Education	2019
3	Electronic Communication Systems	George Kennedy and Bernard Davis	4th Edition	Tata McGraw Hill	2009
Web References					
1.https://logisim.app/					
2.https://www.tinkercad.com/					

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25EC151/25EC251)

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	**Statement**	**Knowledge Level, KL**
CO1	Identify various active and passive components and will learn to work with measuring instruments.	**K1**
CO2	Apply PN Junction diode and Zener diode in few applications.	**K3**
CO3	Implement a switch using BJT and NAND, NOR & NOT logic gates using CMOS.	**K3**
CO4	Realize Boolean expressions using logic gates.	**K3**
CO5	Apply IC 741C and IC 555 in few applications.	**K3**
CO6	Familiarize with Arduino and configure small circuits.	**K3**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	1	2				2		3	2		
CO2	1	1		2	1			3	2		
CO3	1	3	3	3	3		2	3	1		
CO4	2	3	2	2	3		2	3	1		
CO5	3	3	2	3	3		2	3	1		
CO6	1	3	3	3	3	2	2	3	1		
Course Avg.	1.50	2.50	2.50	2.60	2.60	2.00	2.00	3.00	1.33		

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25EC151/25EC251)

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	**100**

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1, CO2, CO3,	Conduction of experiment (5), quiz (5), viva&lab(10)	25%, 25%, 50%	20
PA2	CO4, CO5, CO6	Conduction of experiment (5), quiz (5), viva&lab(10)	25%, 25%, 50%	20
	Attendance			10 Marks
Total Marks				**50**

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Experiment Conduction	CO1, CO2, CO3, CO4, CO5, CO6	25
Viva-Voce	CO1, CO2, CO3, CO4, CO5, CO6	25
Total Marks		**50**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	I/II	Computer Aided Design Lab	25ME152/ 25ME252	0	0	2	1

Teaching & Learning Scheme (in hours per semester)

Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
0	0	24	6	30

Course Objective (s)

The course aims:

1. To understand the principles of engineering drawing including projection, dimensioning, and interpretation.
2. To develop skills in 2D drafting using modern CAD tools.
3. To enable students to produce professional technical drawings for engineering communication.

Syllabus

Unit No.	Topics	CI Hours
1	**Unit-1: Introduction to Basic elements of Engineering Drawing** Introduction to the different drawing instruments and application, Convention of lines and its application (Thick, Thin, Axis etc.,), Practice use of drawing instruments, Scales - Full Scale, Reduced Scale and Enlarged Scale, Lettering & Dimensioning, Construct different polygons.	2
2	**Unit-2: Introduction to Orthographic Projections – Points** Introduction to Projections-Principal Planes of Projection and Principal Views, Introduction to First angle and Third angle method, their symbols, Projection of points in All 4 Quadrants.	2
3	**Unit-3: Introduction to Orthographic Projections - Lines** Projection of lines in First angle, Parallel and Perpendicular to both the planes, Parallel to one plan and Perpendicular to another plan, Parallel to one plan and inclined to another plan.	2
4	**Unit-4: Introduction to Orthographic Projections – Plane Surfaces** Parallel to one plane and Perpendicular to another plane, Planes Perpendicular to one plane and inclined to the other plane, Resting on Edge, Corner, Inclined to HP and VP	2

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5	**Unit-5: Introduction to Computer Aided Drafting** Introduction to AutoCAD, Familiarization of CAD user interface, setting up new drawing, Selecting Various plotting parameters, Introduction to 2D Drafting Commands - Drawing	**4**
6	**Unit-6: Introduction to 2D Drafting Commands - Editing & Modifying** Editing and modify commands, Dimensioning and applying constraints, Introduction to 2D drafting	**4**
7	**Unit-7: Introduction to 3D Modelling** Introduction to 3D Commands in AutoCAD	**4**
Value Added Experiments		
8	Introduction to Solid Works - 2D	**2**
9	Introduction to Solid Works - 3D	**2**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Engineering Drawing	Bhatt N.D.	55th Edition	Charotar Publishing House	2025
2	Engineering Drawing and Graphics + AutoCAD	Venugopal K.	6th Edition	New Age International	2022
3	Engineering Drawing	Gill P.S.	13th Edition	S.K. Kataria & Sons	2016

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Engineering Drawing	Basant Aggarwal	3rd Edition	Tata McGraw-Hill	2019
2	Fundamentals of Engineering Drawing	Luzadder Warren J	10th Edition	Prentice Hall of India Private Limited	1989

Web References

1. Fusion Fundamentals Course (ONLINE), AUTODESK

Certification for the self-learning component should preferably be included in the assessment.

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COURSES OUTCOMES (25ME152/ 25ME252)

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	**Statement**	**Knowledge Level, KL**
CO1	Identify and interpret drawing instruments, line conventions, scales, and lettering for engineering communication.	**K2**
CO2	Construct orthographic projections of points, lines, and planes using engineering drawing principles.	**K3**
CO3	Create and modify 2D drawings using CAD software by applying coordinate systems, constraints, and dimensioning techniques with proper plotting and file management.	**K3**
CO4	Develop basic 3D models of engineering components using CAD software and visualize them effectively.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	2	1			1				2		1
CO2	2	2	1					1	1		
CO3	2	2	2		3			1	2		2
CO4	2	2	3		3			1	2		2
CO5											
Course Avg.	2	1.75	2		2.33			1	1.75		1.67

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25ME152/25ME252)

Evaluation Pattern			
Internal Component (CIA)		**External Component**	**Total Marks**
PA1	**PA2**		
20 +5 (Attendance)	20 +5 (Attendance)	50	**100**

Assessment Plan for CIA				
	CO	**Assessment Tool**	**Remarks**	**Weightage of Marks in CIA**
PA1	**CO1**	Experiment Conduction Quiz Lab Records	Sheet/Drawing Work Online Quiz Continuous Lab Work	5 Marks 5 Marks 10 Marks
	CO2			
	Attendance			5 Marks
PA2	**CO3**	Experiment Conduction Quiz Lab Records	Sheet/Drawing Work Online Quiz Continuous Lab Work	5 Marks 5 Marks 10 Marks
	CO4			
	Attendance			5 Marks
Total Marks				**50**

Assessment Plan for External Component		
Assessment Tool	**COs Coverage**	**Weightage of Marks in External Component**
Experiment Conduction	CO1, CO2, CO3, CO4	25
Viva-Voce	CO1, CO2, CO3, CO4	25
Total Marks		**50**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. TECH	I/II	Digital Manufacturing Lab	25ME151/ 25ME251	0	0	2	1

Teaching & Learning Scheme (in hours per semester)

Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
0	0	24	6	30

Course Objective (s)

The course aims:

1. To familiarize students with fundamental workshop and machine shop practices
2. To introduce students to digital and computer-controlled manufacturing technologies
3. To develop awareness of automation and mechatronic systems
4. To provide exposure to thermal and energy systems in manufacturing/service applications**,**

Syllabus

Unit No.	Topics	CI Hours
1	Introduction to Workshop Practices	**2**
2	Introduction to Machine Shop	**2**
3	Introduction to CNC Machine Shop	**2**
4	Introduction to CNC Wood Router Shop	**2**
5	Introduction to 3D Printing	**2**
6	Introduction to Automation	**2**
7	Introduction to LASER engraving	**2**
8	Introduction to Drone Technology	**2**
9	Introduction to Advance RAC	**2**
10	Value added experiment-1 : Introduction to sheet metal shop	**2**
11	Value added experiment-2: Introduction to Black Smithy shop	**2**

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12	Value added experiment-2: Introduction to Shaper Mchine	**2**

Prescribed Text Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	**Workshop Technology (Vol. I, II & III)**	B.S. Raghuwanshi	2	(Dhanpat Rai & Co.)	2024
2	**Elements of Workshop Technology (Vol. I & II)**	S.K. Hajra Choudhury, A.K. Hajra Choudhury, Nirjhar Roy	3	Media Promoters & Publishers	2024
Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	**Manufacturing Technology (Vol. I & II)**	P.N. Rao	2	Tata McGraw Hill	2018
2	**CNC Machines**	B.S. Pabla, M. Adithan	3	New Age International	2018
3	**Pneumatics and Hydraulics**	Andrew Parr	1	Jaico	1993
Web References					
1. https://nptel.ac.in/courses/112105126					
2. https://nptel.ac.in/courses/112103174					
3. https://all3dp.com					

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25ME151/25ME251)

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	**Statement**	**Knowledge Level, KL**
CO1	Identify and demonstrate the use of workshop tools, safety practices, and the purpose of different shops	**K2**
CO2	Perform basic operations in carpentry, fitting, and welding using appropriate techniques	**K3**
CO3	Interpret and explain the working of conventional machines such as lathe and drilling machines	**K2**
CO4	Analyze the functioning, workflow, and applications of 3D printing and related digital manufacturing technologies	**K3**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	2	1	1	1	3	2	3	1	1		2
CO2	2	1	2		3	2	2	1	1		1
CO3	3	2	2	2	3	2	2	1	1		1
CO4	2	2	2	2	3	2	2	1	1	2	3
Course Avg.	**2.25**	**1.50**	**1.75**	**1.67**	**3.00**	**2.00**	**2.25**	**1.00**	**1.00**	**2.00**	**1.75**

3 – High; 2 – Medium; 1 – Low

ABES Engineering College, Ghaziabad
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EVALUATION PATTERN (25ME151/25ME251)

Evaluation Pattern			
Internal Component (CIA)		**External Component**	**Total Marks**
PA1	**PA2**		
20 +5 (Attendance)	20 +5 (Attendance)	50	**100**

Assessment Plan for CIA				
	CO	**Assessment Tool**	**Remarks**	**Weightage of Marks in CIA**
PA1	**CO1**	Quiz, Internal Examination/Viva, Quiz, Lab Record	25%+25%+50%	20
	CO2			
	Attendance			5 Marks
PA2	**CO3**	Quiz, Internal Examination/Viva, Lab Record	25%+25%+50%	20
	CO4			
	Attendance			5 Marks
Total Marks				**50**

Assessment Plan for External Component		
Assessment Tool	**COs Coverage**	**Weightage of Marks in External Component**
Experiment Conduction	CO1, CO2, CO3,CO4	25
Viva-Voce	CO1, CO2, CO3, CO4	25
Total Marks		**50**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech	I	Web Designing Workshop - I	25VA151	0	0	3	2

Teaching & Learning Scheme (in hours per semester)

Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
0	0	36	54	90

Course Objective (s)

The course aims:

1. Introduce students to the fundamentals of web technologies and the full-stack development environment.
2. Familiarize students with modern tools like VS Code and Git for development and version control practices.
3. Provide hands-on experience with HTML, CSS, and GitHub for collaborative development.
4. Develop practical skills in creating structured, styled, and interactive web pages.

Syllabus

Unit No.	Topics	CI Hours
1	**Introduction to Web Technology, Full Stack Development**: Web Designing, Web Development, Client Server Architecture, Introduction to Text Editors, Installation & Set up of VS Code Editor, Introduction of Git & GitHub, Installation & Configuration of Git Account, Introduction to HTML & HTML Features, HTML Page Structure, Elements in HTML (Semantic and Non-Semantic), Types of Elements (Inline/Block), Text formatting: headings, paragraphs, HTML Lists, Unordered Lists, Ordered Lists, Description Lists, Nesting and Mixed Lists, Lists for Navigation Menus, HTML links, Images and attributes, Comment in HTML, HTML Formatting Elements / Special Character & Symbol, Text related Elements (Quotation and Citation Elements), Using Multimedia (Image/Video/Audio/YouTube) in HTML, Creating & Using Hyperlinks, Accessibility and Semantic Considerations, Practical	**10**
2	**Web Page Development using HTML**: HTML Table: Creating and using Tables, Basic Table Structure, Table Headers and Captions, HTML Table: Rowspan and Colspan, Table Attributes, Responsive Tables, HTML Table: Nested Tables, Accessibility in Tables, Practical, HTML Form: The	**8**

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	<form> Element, Input Elements, Label and Fieldset, HTML Form: Select and Option Tags, TextArea, Button Element, HTML Form: Form Attributes, Input Validation, Accessibility and Usability, What is Git and why use it?, Git setup and config (git config), Creating a local repository: git init, Tracking changes: add, commit, status, log, .gitignore basics, Practical	
3	**Version Control with Git & GitHub and CSS Fundamentals :**Creating a GitHub account, Connecting local repos to GitHub: remote add, push, Cloning and pulling repositories, GitHub repository structure and README.md, Basic collaboration terms: fork, pull request, branch (intro), Branching: git branch, git checkout, git switch, Merging branches, handling merge conflicts, GitHub collaboration workflow, Best practices for commits, branches, and messages, Practical with all steps of git, CSS selectors: element, class, ID, Color, fonts, text alignment, font size, Box model: padding, margin, border, Box model: padding, margin, border, Applying styles: inline, internal, external	**9**
4	**CSS for Advanced Styling and Responsive Layout:** display: block, inline, inline-block, float and clear, Introduction to Flexbox: justify-content, align-items, flex-direction, Practical in CSS, Units: px, %, em, rem, vh, vw, Media queries and breakpoints, Media queries and breakpoints, Mobile-first design, Responsive navigation concepts, Practicals, Project planning and design, Version control in large projects, Deploy using GitHub Pages, Final portfolio polishing, Project using HTML & CSS	**9**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Learning React: Modern Patterns for Developing React Apps	Alex Banks & Eve Porcello	2nd Edition	O'Reilly Media	2020
2	Full Stack Development with Spring Boot and React	Juha Hinkula	4th Edition	Packt Publishing	2023
3	Tailwind CSS	Noel Rappin	1st Edition	O'Reilly Media	2024

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Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	JavaScript: The Good Parts	Douglas Crockford	1st Edition	O'Reilly Media	2008
2	HTML, Bootstrap, CSS, Tailwind, & Cordova: A Complete Guide	Chong Lip Phang	1st Edition	Independently Published	2023
3	Designing Applications with Spring Boot 2.2 and React JS	Dinesh Rajput	1st Edition	BPB Publications	2020

Web References
1. https://www.coursera.org/learn/introduction-html-css-javascript
2. https://developers.google.com/profile/badges/playlists/webdev/learn-html
3. https://www.coursera.org/learn/react-basics
4. https://www.edx.org/learn/node-js/ibm-back-end-application-development-with-node-js-and-express

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25VA151)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Demonstrate understanding of basic web technologies and client-server architecture.	**K2**
CO2	Develop well-structured web pages using HTML elements, forms, and multimedia integration.	**K3**
CO3	Apply Git and GitHub for version control and collaborative web development.	**K3**
CO4	Design CSS for styling, layout design, and responsive formatting.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	1		2				1		1
CO2	3	2	2		2			1	2		1
CO3	1	1	1		3		2	3	2	2	2
CO4	3	2	3		2			1	2		1
Course Avg.	2.5	1.75	1.75		2. 25		2	1. 67	1.75	2	1. 25

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25VA151)

Evaluation Pattern			
Internal Component (CIA)		**External Component**	**Total Marks**
PA1	**PA2**		
20 +5 (Attendance)	20 +5 (Attendance)	50	**100**

Assessment Plan for CIA				
	CO	**Assessment Tool**	**Remarks**	**Weightage of Marks in CIA**
PA1	CO1	Lab Test / Practical File Evaluation / Quiz	Continuous evaluation – Units 1 & 2	20
	CO2			
PA2	CO3	Lab Test / Practical File Evaluation / Quiz	Continuous evaluation – Units 3 & 4	20
	CO4			
	Attendance			10 Marks
Total Marks				**50**

Assessment Plan for External Component		
Assessment Tool	**COs Coverage**	**Weightage of Marks in External Component**
Experiment Conduction	CO1, CO2, CO3, CO4	25
Viva-Voce	CO1, CO2, CO3, CO4	25
Total Marks		**50**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech	1st	C++ Programing Lab	25CS151	0	0	2	1

Teaching & Learning Scheme (in hours per semester)

Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
0	0	24	6	30

Course Objective (s)

The course aims:

1. To introduce students to the fundamental concepts of problem solving through the development of algorithms and flowcharts.
2. To provide a strong foundation in basic C++ programming constructs such as data types, operators, and control statements.
3. To develop the ability to apply iterative structures and arrays for solving computational problems in C++.
4. To promote modular programming skills by enabling students to design and implement functions for structured problem solving.
5. To familiarize students with string manipulation techniques and essential C++ features required for solving text-based problems.

Unit No.	Topics	CI Hours
1	Draw the Flowchart and write an Algorithm to check whether the user input year is Leap year or not.	**2**
2	Write a program in C++ to find the largest of three numbers.	**2**
3	Write a program in C++ to determine the grade of a student based on marks.	**2**
4	Write a program in C++ to calculate the factorial of a number.	**2**
5	Write a program to multiply two matrices and display the resultant matrix.	**2**
6	Write a program in C++ to calculate the factorial of a number using a function.	**2**
7	Write a program in C++ to check whether a number is prime using a function.	**2**
8	Write a program to swap two integers using the concept of call by value and call by reference.	**2**

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9	Write a program in C++ to count the number of vowels in a given string.	**2**
10	Write a program in C++ to check whether the given string is a palindrome or not.	**2**
	Value added experiments	
1	Write a program in C++ to find an element in array using Binary Searching technique.	**2**
2	Write a program in C++ to sort an array using Bubble Sort algorithm.	**2**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	C++ Primer	Stanley B. Lippman, Josée Lajoie, Barbara E. Moo	5th Edition	Addison-Wesley Professional	2012
2	Object-Oriented Programming with C++	E. Balagurusamy	8th Edition	McGraw Hill Education (India) Private Limited	2021

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	C++: The Complete Reference	Herbert Schildt	4th Edition	McGraw-Hill Education (Osborne/McGraw-Hill; Tata McGraw-Hill in India)	2003
2	Object-Oriented Programming in C++	Robert Lafore	4th Edition	SPearson Education India / SAMS	2008

Web References

1. https://www.programiz.com/cpp-programming
2. https://www.geeksforgeeks.org/c-plus-plus
3. https://leetcode.com
4. https://www.codechef.com

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25CS151)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Understand flowcharts and algorithms for common computational problems and translate them into basic C++ program structures.	**K2**
CO2	Apply appropriate data types, operators, and control statements to design and implement simple to moderately complex C++ programs.	**K3**
CO3	Use iterative constructs and arrays to write efficient C++ programs for solving computational and data-processing tasks.	**K3**
CO4	Implement modular programming by decomposing problems into functions and applying parameter passing, return types, and functional abstraction in C++.	**K3**
CO5	Analyze and process strings in C++ using built-in and custom methods to solve pattern-based and text-manipulation problems.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	1	1	2				3		1
CO2	3	3	3	2	3						
CO3	3	3	3	2	3						2
CO4	3	3	3	3	3						
CO5	3	3	2	2	3						2
Course Avg.	3.00	2.80	2.40	2.00	2.80				3.00		1.67

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25CS151)

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	**100**

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Conduction Of Program/Quiz	100%	20
	CO2			
PA2	CO3	Conduction Of Program/Quiz	100%	20
	CO4			
	CO5			
	Attendance			10 Marks
Total Marks				**50**

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Experiment Conduction	CO1, CO2, CO3,CO4,CO5	25
Viva-Voce	CO1, CO2, CO3, CO4, CO5	25
Total Marks		**50**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech	II	Applied Calculus and Differential Equation	25AS202	2	1	0	3

Teaching & Learning Scheme (in hours per semester)

Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
42	12	0	36	90

Course Objective (s)

To enable students to acquire core mathematical competencies in calculus and ordinary differential equations, enabling the systematic application of mathematical methods to model, analyze, and solve engineering and real-world problems.

Syllabus

Unit No.	Topics	CI Hours
1	**Unit 1: Multivariable Differential Calculus** Function of two variables, Partial differentiation, Homogeneous functions, Euler's theorem, Total derivative, Jacobian and its applications, Maxima and minima of functions of two variables.	8
2	**Unit 2: Ordinary Differential Equations** nth-order linear differential equations with constant coefficients, Linear Differential equations with variable coefficients, Solution of Homogeneous L.D.E. (Euler-Cauchy equation), Solution of Non-Homogeneous L.D.E (Second Order) by changing independent variable, Method of variation of parameters, Application of differential equations in solving engineering problems.	9
3	**Unit 3: Laplace Transform and Applications** Laplace transform: Definitions and properties, Laplace transforms of derivatives and integrals, Evaluation of integrals by Laplace transform, Unit step function, Laplace transform of Periodic functions, Inverse Laplace transform, and Convolution theorem, Application of Laplace Transform to solve ordinary differential equations.	9
4	**Unit 4: Multiple Integrals and Special Functions** Double and triple integrals, Change of order of integration, Change of variables. Introduction to Beta, Gamma functions and their properties, Dirichlet's integral and its applications to area and volume	8

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5	**Unit 5: Vector Calculus** Gradient, Divergence and Curl, Directional derivatives, Line integral, Surface integral, Volume integral, Gauss's Divergence theorem, Green's theorem, Stoke's theorem (without proof) and their applications.	**8**

Prescribed Text Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	B V Ramana	Higher Engineering Mathematics	1st	Tata McGraw-Hill	2012
2	Higher Engineering Mathematics	B.S. Grewal	42th	Khanna Publishers	2012
3	Advanced Engineering MAthematics	Jain R.K., Iyengar S.R.	6th	New Age International Publishers	2014
Reference Books					
S. No.	**Book Title**	**Authors**	**Edition**	**Publisher**	**Year**
1	Advanced Engineering Mathematics	Erwin Kreyszig	10th	John Wiley & Sons	2010
2	Calculus and Analytical Geometry	G.B Thomas, R L Finne	9th	Pearson	2002
3	Advance Engineering Mathematics	Peter V. O'Neil	8th	Thomson (Cengage) Learning	2007
4	Laplace Transforms by Schaum's	Spiegel	2005 Edition	McGraw Hill Education	2005

Web References
1. https://en.wikipedia.org/wiki/Jacobian_matrix_and_determinant
2. https://en.wikipedia.org/wiki/Laplace_transform
3. https://en.wikipedia.org/wiki/Stokes%27_theorem

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25AS202)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Apply the concepts of partial differentiation, Jacobians, and optimization techniques to analyze and solve multivariable engineering problems.	**K3**
CO2	Apply the concepts of differentiation to solve nth order ordinary differential equations.	**K3**
CO3	Use Laplace transform techniques to solve ordinary differential equations arising in engineering problems.	**K3**
CO4	Understand the concepts of multiple integrals, Beta and Gamma functions, and apply Dirichlet's integral to determine area and volume.	**K3**
CO5	Use vector calculus concepts to solve problems involving directional derivatives, line integrals, surface integrals, and volume integrals.	**K3**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	1	2	1	1	2				1		1
CO2	3	2	1	1	3				1		2
CO3	3	2	1	1	3				1		2
CO4	1	2		1	2				1		2
CO5	3	2	1	1	3				1		2
Course Avg.	2.20	2.00	1	1.00	2.60				1.00		1.80

3 – High; 2 – Medium; 1 – Low

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Estd. 2000

EVALUATION PATTERN (25AS202)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term - I	CO1(10 marks) +CO2 (5 marks)	15
Mid Term -II	CO2(5 marks) +CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech	2nd	Data Structure Using C++	25CS201	3	0	0	3

Teaching & Learning Scheme (in hours per semester)

Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
36	0	0	54	90

Course Objective (s)

The objective of this course is to provide students with a strong foundation in data structures and algorithms, enabling them to understand, implement, and analyze fundamental data structures, sorting and searching techniques, hashing, and STL-based structures with appropriate time and space complexity analysis for efficient problem solving.

Syllabus

Unit No.	Topics	CI Hours
1	**Introduction:** Data Structures and its types importances,primitives operations on data structures. Basic Terminology, Elementary Data Organization, Built in Data Types in C++.Algorithm, Efficiency of an Algorithm, Time and Space Complexity, Asymptotic notations:Big Oh, Big Theta and Big Omega, Time-Space trade-off. Abstract Data Types (ADT). **Searching:** Concept of Searching, Sequential search,Binary Search. computation of Time and space complexity.	6
2	**Sorting:** Bubble-Sort ,Insertion-Sort, Selection-Sort and its C++ implementation and Time complexity analysis. Sorting in Linear time: Counting-sort,key characteristics of counting sort, C++ implementation ,algorithm and time complexity analysis. Radix Sort, Bucket Sort .Comparision of sorting techniques in terms of Best, Average and worst time complexity analysis.	8
3	**Linked lists:** Array Implementation and Pointer Implementation of Singly Linked Lists, Doubly Linked List, Circularly Linked List, Operations on a Linked List. Insertion, Deletion, Traversal and its Timecomplexity analysis (Worst,Average,Best), Polynomial Representation and Addition Subtraction & Multiplications of Single variable & Two variables Polynomial.	8

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4	**Hashing:** introduction of hashing and its need.Definition , hash table, Hash function: Division Modulo Method,Digit Extraction,Mid Square, Folding,Custom Hash Function.Operations Over Hash Table: Search, Insertion,Deletion. **Collision:** Illustration of Collision,Collision Resolution Techniques:Open Hashing or Separate Chaining, Closed Hashing or Open Addressing: Linear Probing, Quadratic Probing, Double Hashing. Programming based hashing technique	**6**
5	**Set:** Ordered and Unordered Set,Definition and their difference,Operations on set(STL): insert (), erase (), size (), equality check of two set, includes (), find(), count(), accumulate (). union of sets, intersection of sets, Difference of sets, sorting a set. Time complexity comparision of different operation for ordered and unordered sets. **Map:** Ordered and unordered map: Definition of maps, key-value concept, Operations on map: nsert(), erase(), size(),looping in map, sorting: by value, by key,custom sorting	**8**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Data Structures Using C++	Reema Thareja	2nd Edition	Oxford University Press	2014
2	Data Structures and Algorithms in C++	Adam Drozdek	4th Edition	Cengage Learning	2012

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Data Structures Using C and C++	Yedidyah Langsam, Moshe J. Augenstein, Aaron M. Tenenbaum	2nd Edition	Pearson (Indian Edition)	2015
2	Data Structures and Algorithms in C++	Michael T. Goodrich, Roberto Tamassia, David M. Mount	2nd Edition	John Wiley & Sons (Wiley)	2011

Web References

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1. https://www.programiz.com/
2. https://cp-algorithms.com/
3. https://leetcode.com/
4. https://www.geeksforgeeks.org/

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25CS201)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Understand fundamental data structures, algorithms, searching techniques, and analyze their time and space complexities using asymptotic notations.	**K2**
CO2	Understand, implement, and analyze various sorting algorithms based on their time complexity	**K3**
CO3	Understand, implement, and analyze linked list data structures and perform polynomial representation and operations (addition, subtraction, and multiplication) using suitable data structures with time-complexity analysis.	**K3**
CO4	Understand and implement hashing techniques, hash functions, collision resolution methods, and perform search, insertion, and deletion operations in hash tables using C++.	**K3**
CO5	Understand and use ordered and unordered sets and maps in C++, perform standard STL operations, apply set and map algorithms, analyze time complexities, and implement sorting using keys, values, and custom criteria.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	2	3	2				2		1
CO2	3	1	3	3	3				2		
CO3	3	1	3	3	3				2		
CO4	3	3	3	3	2				1		
CO5	3	2	3	3	3				2		2
Course Avg.	3.00	1.80	2.80	3.00	2.60				1.80		1.50

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25CS201)

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz / Assignment/ Class Test	100%	4
CO2	Quiz / Assignment/ Class Test	100%	4
CO3	Quiz / Assignment/ Class Test	100%	4
CO4	Quiz / Assignment/ Class Test	100%	4
CO5	Quiz / Assignment/ Class Test	100%	4
	Attendance	100%	10
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term - I	CO1(10 marks) +CO2 (5 marks)	15
Mid Term -II	CO2(5 marks) +CO3 (10 marks)	15
End Term	CO1+CO2+ CO3+CO4+ CO5 (4+4+4+14+14 marks)	40
Total Marks		70

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech	II	Web Designing Workshop - II	25VA251	0	0	3	2

Teaching & Learning Scheme (in hours per semester)

Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
0	0	36	54	90

Course Objective (s)

The course aims:

1. To enable students to design and develop responsive, mobile-first web pages using Bootstrap and Tailwind CSS with effective use of grid systems, utility classes, and UI components.
2. To develop students' ability to create dynamic and interactive web functionality using JavaScript basics, DOM manipulation, event handling, and API communication using modern asynchronous techniques
3. To equip students with the skills to build modular, reusable, and state-managed front-end applications using React components, hooks, routing, and data handling.
4. To provide students with the capability to build backend services and RESTful APIs using Java and Spring Boot by applying OOP concepts and implementing CRUD operations.

Syllabus

Unit No.	Topics	CI Hours
1	**Responsive Web Design (Bootstrap & Tailwind CSS) :**Introduction to Web Design Frameworks; Bootstrap Essentials: Breakpoints, Responsiveness, Typography, Images, Buttons, Badges, Carousel, Progress Bars, Modals, Grid layout system (rows/columns), Forms, Navbars, Cards, Bootstrap utility classes & customization; Tailwind CSS Essentials: Installing Tailwind via CDN & build tools, Utility classes for layout, spacing, typography, color, Responsive design & breakpoints, Customizing themes, using @apply.	8
2	**JavaScript Programming (Basics to APIs**): JavaScript Basics: Variables, Data Types, Operators, Loops and Conditionals, Arrays, Objects, Functions, DOM Manipulation & Event Handling, JSON & working with JSON data, for...in, for...of loops; Intermediate JavaScript & APIs: Arrow Functions, map, filter, reduce, Promises, async/await, Fetch API, Error handling, Local Storage usage.	9
3	**React Fundamentals** : React Introduction: Components, JSX, Props, useState Hook, Event handling & State Management; Working with Data in React: Lists rendering, Keys, Controlled Forms, useEffect (fetch data & side effects), Lifting	11

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	state up (parent-child communication); Routing & Component Structure: React Router DOM, Navigation links, layout components, Code organization & reusable components; Styling in React: Using Bootstrap or Tailwind inside React, Component reuse, Error handling & loading states, Deployment (Netlify or Vercel).	
4	**Java Backend & API Development with Spring Boot:** Java OOP Concepts, Streams and Lambda expressions; Introduction to Spring Boot: Spring Initializer, Auto-configuration, Controller vs RestController; REST API Implementation: CRUD operations (Add, Update, Delete, Search).	**8**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Learning React: Modern Patterns for Developing React Apps	Alex Banks & Eve Porcello	2nd Edition	O'Reilly Media	2020
2	Full Stack Development with Spring Boot and React	Juha Hinkula	4th Edition	Packt Publishing	2023
3	Tailwind CSS	Noel Rappin	1st Edition	O'Reilly Media	2024

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	JavaScript: The Good Parts	Douglas Crockford	1st Edition	O'Reilly Media	2008
2	HTML, Bootstrap, CSS, Tailwind, & Cordova: A Complete Guide	Chong Lip Phang	1st Edition	Independently Published	2023
3	Designing Applications with Spring Boot 2.2 and React JS	Dinesh Rajput	1st Edition	BPB Publications	2020

Web References

1. https://www.coursera.org/specializations/html-css-javascript-for-web-developers
2. https://www.freecodecamp.org/learn/responsive-web-design-v9/
3. https://alison.com/course/responsive-web-development-with-bootstrap-4
4. https://www.mygreatlearning.com/academy/learn-for-free/courses/introduction-to- web-designing
5. https://www.udemy.com/topic/web-design/

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Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25VA251)

Course Outcomes (COs):
Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Explain the architecture, core modules, and event-driven programming model of Node.js for building scalable server-side applications.	**K2**
CO2	Apply JavaScript and API techniques to develop interactive and responsive web pages.	**K3**
CO3	Analyze and build dynamic user interfaces using React components, state, hooks, and routing.	**K4**
CO4	Analyze and implement RESTful web services using Spring Boot by applying JAVA OOP concepts, Streams, Lambda expressions, controllers, and CRUD operations.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	2		3				1		3
CO2	3	2	2	2	3				1		3
CO3	3	3	3	2	3			1	2	2	3
CO4	3	3	3	2	3			1	2		3
Course Avg.	3.00	2.50	2.50	2.00	3.00			1.00	1.50	2.00	3.00

3 – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25VA251)

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	**100**

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Lab Test / Practical File Evaluation / Quiz	Continuous evaluation – Units 1 & 2	20
	CO2			
PA2	CO3	Lab Test / Practical File Evaluation / Quiz	Continuous evaluation – Units 3 & 4	20
	CO4			
	Attendance			10 Marks
Total Marks				**50**

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Experiment Conduction	CO1, CO2, CO3, CO4	25
Viva-Voce	CO1, CO2, CO3, CO4	25
Total Marks		**50**

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Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech	II	DSA Lab Using C++	25CS251	0	0	2	1

Teaching & Learning Scheme (in hours per semester)

Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)	Total no. of Hours per Semester
L	T	P	SL*	
0	0	24	6	30

Course Objective (s)

The course aims:

1. To provide the strong foundation in data structures and algorithms, enabling them to understand, time and space complexity.
2. To provide a strong foundation in data structure to understand sorting and searching techniques.
3. To develop the ability to apply constructors to create a Node to implement Linked list data structure and its types.
4. To promote hashing, and STL-based structures for problem solving in data structures.
5. To understand how to keep unique data in data structures using SET. And also keep the data using Key for the MAP data structure.

Unit No.	Topics	CI Hours
1	Write a C++ program to perform Linear Search on a given list of elements and display whether the element is found or not along with its position.	2
2	Write a C++ program to perform Binary Search on a sorted array and analyze its time complexity.	2
3	Write a C++ program to sort an array using Bubble Sort and display the sorted elements.	2
4	Write a C++ program to implement Selection Sort and compare its best, average, and worst-case time complexity.	2
5	Write a C++ program to create a Singly Linked List and perform insertion and traversal operations.	2
6	Write a C++ program to delete a node from a linked list and display the updated list.	2

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7	Write a C++ program to implement a Hash Table using Linear Probing for collision resolution.	**2**
8	Write a C++ program to perform search operation in a hash table using open addressing.	**2**
9	Write a C++ program to demonstrate Set operations such as insert, delete, find, and display elements.	**2**
10	Write a C++ program to implement a Map and perform key-value operations with sorting.	**2**
	Value added experiments	
1	Write a C++ program to Implement a basic hashing system for password verification.	**2**
2	Write a C++ program to Store, search, and sort student records using Map and Set.	**2**

Prescribed Text Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	C++ Primer .	Stanley B. Lippman, Josée Lajoie, Barbara E. Moo	5th Edition	Addison-Wesley Professional	2012
2	Object-Oriented Programming with C++	E. Balagurusamy	8th Edition	McGraw Hill Education (India) Private Limited	2021

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	C++: The Complete Reference	Herbert Schildt	4th Edition	McGraw-Hill Education (Osborne/McGraw-Hill; Tata McGraw-Hill in India)	2003
2	Object-Oriented Programming in C++	Robert Lafore	4th Edition	SPearson Education India / SAMS	2008

Web References

1. https://www.programiz.com/cpp-programming
2. https://www.geeksforgeeks.org/c-plus-plus
3. https://leetcode.com
4. https://www.codechef.com

Certification for the self-learning component should preferably be included in the assessment.

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COURSE OUTCOMES (25CS251)

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	**Statement**	**Knowledge Level, KL**
CO1	Students will understand the basic concepts of searching techniques such as linear and binary search.	**K2**
CO2	Students will be able to implement and analyze different sorting algorithms including Bubble, Selection sort.	**K3**
CO3	Students will learn to design and perform primitive operations on linked lists.	**K3**
CO4	Students will understand hashing techniques, hash functions, collision resolution methods, and implement hashing using programming.	**K3**
CO5	Students will be able to analyze and implement STL-based Set and Map for efficient data storage, retrieval, and sorting operations.	**K4**

KL- Bloom's Knowledge Level (K_1, K_2, K_3, K_4, K_5, K_6)
K_1- Remember, K_2- Understand, K_3- Apply, K_4- Analyze, K_5- Evaluate, K_6- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	2	3	2				3		1
CO2	3	3	3	3	3				3		2
CO3	3	3	3	3	3				3		2
CO4	3	3	3	3	2				2		
CO5	3	3	3	3	3				3		3
Course Avg.	3.00	2.80	2.80	3.00	2.60				2.80		2

4. – High; 2 – Medium; 1 – Low

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EVALUATION PATTERN (25CS251)

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	**100**

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Conduction Of Program/Quiz	100%	20
	CO2			
PA2	CO3	Conduction Of Program/Quiz	100%	20
	CO4			
	CO5			
	Attendance			10 Marks
Total Marks				**50**

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Experiment Conduction	CO1, CO2, CO3,CO4,CO5	25
Viva-Voce	CO1, CO2, CO3, CO4, CO5	25
Total Marks		**50**